Çukurova University

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CEN-118 Algorithms and Programming Lab. II

GROUP NO: 23

CANDY CRUSH GAME PROJECT REPORT

FRUIT RUSH

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Abstract

Our game *"Fruit Rush"* is a prototype inspired by the game "Candy Crush Saga". It is developed in C language, utilizing the Raylib library for graphical operations.

Introduction

With its colorful graphics, sweet music, and unique style, "Fruit Rush" aims to provide users with an enjoyable gaming experience reminiscent of "Candy Crush Saga". "Fruit Rush" features levels with various tasks, similar to "Candy Crush Saga", which must be completed to progress. Each level requires the completion of specific tasks, such as achieving certain scores within a limited number of moves or within time constraints. For instance, players may need to create a certain number of items through special matches or reach a target score within a given time frame.

Introduction

In *"Fruit Rush"*, four different fruits represent the classic four different candies of "Candy Crush Saga". Players score points by aligning three, four, or five of the same fruits in a row or column. While minimum points are awarded for three-match combinations, maximum points are awarded for five-match combinations, triggering the creation of a special item called a *"cookie"*. This item, similar to the special candies in "Candy Crush Saga", provides advantages by destroying surrounding fruits. Another unique feature of "Fruit Rush" is the "cookie's" ability to destroy adjacent fruits after a certain period. The game includes start, end, win, and lose screens, as well as on-screen counters for score, moves, and time to enhance the user experience. "Fruit Rush" also tracks *"high scores"* at certain levels, allowing players to compete with themselves and others. Despite these features, "Fruit Rush" is still a game in development with room for improvement to make it more similar to "Candy Crush Saga".

Methodology

Initially, we formed our group at the beginning of the term. Before the games were announced, we started preliminary research on the Raylib library based on our teachers' advice, which informed us about the technical infrastructure and allowed us to lay a solid foundation for our project. Once the game was announced, we intensified our research. Each group member took an active role, managing the process both face-to-face and online, which significantly helped us manage the process better. This allowed us to conduct extensive coding and debugging sessions.

Core Algorithm

The first algorithm forming the basis of our project was the "Core Algorithm", a product of our preliminary research. This algorithm included the game's basic dynamics and matching mechanism. Throughout the process, all team members took roles in developing different aspects of this algorithm. Ilayda led the establishment and development of the algorithm's dynamics, while Melisa supported Ilayda by researching new developments. Sude managed the debugging processes, divided the code into different files, and configured the Raylib library. By the end of this process, we developed a game code with mechanics similar to "Candy Crush Saga" but in a more primitive form. The algorithm effectively identified and processed matches on the game board. This initial version of the code provided a solid foundation for subsequent project phases.

Customization Process

After the initial work on the core algorithm and basic code structure, we decided to add innovative features unique to our game. We developed a special feature triggered when five fruits align horizontally or vertically. Initially called the "black square" (due to using Raylib's pre-existing colors before starting the design process), this feature was later named *"cookie"*. This new feature, though similar to the five-match explosion in "Candy Crush Saga", is not identical; the resulting "cookie" clears a 3x3 area around it.

Customization Process

This unique mechanism adds originality to our game. Ilayda led the algorithm development of this new feature, Melisa worked on increasing and enriching the game's levels, and Sude corrected errors during code integration. As a result, the "cookie" feature was successfully integrated, and the game operated smoothly with new levels.

Creating Atmosphere and Level Tasks

To enhance the user experience, we focused on adding atmospheric elements and defining level tasks. The first step was to add sound effects suitable for the game's overall theme. Attention was paid to copyright issues in sound selection. Melisa handled the selection and integration of appropriate sound files. Ilayda defined specific tasks for each game level, determining the challenges and goals players would encounter and integrating them into our code. Sude organized the game's file structure and addressed encountered errors, strengthening the technical infrastructure. These efforts aimed not only to provide a gaming experience for players but also to improve the game's fluidity. As a result of these improvements, our game became more interactive, and sound effects and level tasks enhanced the enjoyment of playing "Fruit Rush".

Design Creation and Code Integration Process

From the beginning, one of our priorities was game design, which we brought to life in this stage. We worked intensively to create visual and auditory elements to enrich the user experience. Sude was directly involved in the creation of the fruit designs, cookie designs, visual atmosphere, and all other designs in Fruit Rush, as well as their integration into the code. The name Fruit Rush also emerged during this process with Sude and was accepted by the group. Melisa and Ilayda organized the game levels, diversified gameplay, and ensured the challenges presented to users were optimal. Additionally, they reviewed and corrected minor errors to ensure the game ran stably and smoothly. We understood the importance of this, especially for a project with numerous variables and dynamic elements, but we also recognized that unused code accumulated after each update. By the end of this process, our game had a rich visual and auditory structure and a robust technical foundation.

Continuation of the Design Process, Code Simplification, and Error Correction:

Design and coding efforts continued as we moved towards the final version of the game. During this stage, Sude deepened her work on visual designs, further enhancing the game's aesthetics.

Continuation of the Design Process, Code Simplification, and Error Correction:

In parallel, Melisa focused on making the code base more efficient by cleaning up unused or outdated code blocks from previous updates, resulting in cleaner and more readable code. This improved the game's performance and made maintenance easier. Ilayda took on the task of identifying and correcting new errors that arose from continuous updates. This three-pronged process represented the culmination of an ongoing development journey from the project's beginning, with each team member working for the game's best possible outcome.

Feedback Process

An important phase of our project development was obtaining feedback. We sought the opinions of Ass.Prof.Dr. Barış Ata, who taught our algorithms class, to provide an external perspective on our project. Dr. Ata noted that the game's speed was too high for players, making it difficult to follow the matching processes. He also suggested adding a "high score" system as a competitive element. Based on this feedback, we initiated a new development process to add animations/visual movements and a high score system to improve player experience. These two significant updates were critical steps to make our game more user-friendly and interactive. Implementing these adjustments allowed us to understand how the game was perceived by the end user and provided an opportunity to enhance the overall quality of the game, adopting a user-centric development approach.

Animation and High Score Addition Process

Following the feedback, we worked on adding animations and a high score system. This process was filled with technical challenges, particularly the integration of animations, which encountered various problems until the end of the development process. The limited time given for the project, due to intense work, was initially agreed to be reported as a deficiency. However, Sude, without giving up, managed to integrate the user-based animation/movement into the game by the end of this process. Thus, the matching speed in the game became dependent on the player. The feedback received was also successfully completed. Ilayda developed and successfully integrated the high score system to measure player performance and create a competitive environment. Melisa focused on sound effects, level design, algorithm development, and debugging during this process, contributing to the game's multifaceted development. As a result, the added animation/movement made in-game actions smoother and visually satisfying. Additionally, the high score system made the player experience more meaningful and competitive. This process significantly enhanced the project's technical capabilities and user interaction.

Final Code Review and Evaluation Process

In the final stage of our project, we tested our game multiple times, meticulously correcting small errors that emerged, aiming to perfect the final code. As a team, we intensely focused on these fine-tuning and optimization efforts. After each test session, we carefully reviewed the code to improve in-game experiences. The development efforts continued with Sude until the end of the project. Meanwhile, Melisa and İlayda worked to optimize the code to its best version. However, some issues could not be fully resolved within the limited project time. Despite slowing down the matching speed with animations, automatically incoming fruits and automatic matches still occurred too quickly, making it difficult for users to follow the game easily. Additionally, occasional excessive waiting times occurred during intermediate screen transitions. Although these issues indicated the need for further development, all team members agreed that the game was ready to be presented in its current form and decided to report these issues.

Reporting and Video Process

After deciding that the code was ready for presentation, we moved on to the reporting and video preparation processes. During the reporting phase, each component of the project was thoroughly examined, and the information was systematically reported under important headings such as the project's scope, methodologies used, results obtained, and challenges encountered.

Reporting and Video Process

Sude played a role in consolidating, organizing, and meticulously preparing the template for the reporting process. In the video process, we used notes taken from the code's significant sections to detail the game's operation and highlight its features. Sude also played a role in combining the video, editing it and transferring it to YouTube. Thus, our project "Fruit Rush" was completed.

Conclusion and Evaluation

PROJECT ACHIEVEMENTS

- **Our game "Fruit Rush" is presented as a prototype of "Candy Crush Saga", resembling the gameplay objectives and mechanics of the referenced game.**
- **Our game features four different fruits and a special "cookie" that appears only with five-match combinations.**
- **In our game, fruits explode appropriately when matched in threes, fours, or fives.**
- **Our game includes a level system.**
- **The first three levels of our game have different tasks, and the subsequent two levels continue with score increases and a decrease in the number of moves.**
- **Our game has a scoring system, and certain level tasks can only be completed by achieving these scores.**
- **A "high score" is tracked in the second level of our game.**
- **Our game includes a move counter, and if the user exhausts the move limit specific to that level, the level cannot be completed.**
- **Various sounds are featured throughout different parts of our game. There is background music that plays when the game is opened, a sound effect when dragging a fruit for matching, and a different sound effect when the "cookie" appears.**

Conclusion and Evaluation

PROJECT ACHIEVEMENTS

- Our game includes a start screen, an interstitial screen designed for level transitions, a screen displayed when a level is not completed, a different interstitial screen when a "high score" is achieved in the second level, and an exit screen when all levels are finished.
- Each different fruit in our game has a unique design, enhancing the visual appeal of the game.
- The start screen, interstitial screens, exit screen, and level backgrounds are all designed with different styles.
- Control buttons are included in our game. These control sound input-output, level transitions, level restarts, and game exit.
- There is an animation/movement when matching fruits, ensuring that the match speed depends on the player.
- Throughout the entire project, all team members maintained constant communication, collaboratively managing the project, and actively participating in all processes.
- Cooperation was essential for all tasks, with everyone striving to fulfill their roles and assist one another.
- Direct communication with instructors was established to improve the project, with feedback being highly valued.

Conclusion and Evaluation

PROJECT SHORTCOMINGS

- **Fruits explode quickly in our game, and sometimes automatically exploding fruits can be hard to follow.**
- **Randomly incoming fruits from top to bottom can sometimes be hard to follow.**
- **There can be occasional waiting issues during screen transitions.**

FUTURE PLANS

- **The explosion speed of fruits will be slowed down, making it easier to follow their movements.**
- **Beautiful animations will be added for fruits and matches to enhance visuals.**
- **More levels will be added to the game, with more varied tasks assigned to different levels.**

Suggestions

- More levels can be added to the game, with varied tasks assigned to these levels.
- Currently, there are four different fruits in the game. This number can be increased.
- Currently, there is one special "cookie" that appears due to a specific condition. The number of special conditions can be increased, with appearances based on specific criteria.
- More sounds can be added to the game.
- The movements of automatically matching fruits and those incoming from the top can be made visible on the screen, making it clearer where the fruits are coming from.
- The explosion speed of the fruits in the game can be slowed down.

References for all Project

Source 1	https://www.adobe.com/express/templates/report
Source 2	https://www.youtube.com/watch?v=-F6THkPkF2I&t=1365s&pp=ygUUYW5kcmV3IGhhbW1lbCByYXlsaWI%3D
Source 3	https://www.youtube.com/watch?v=jWo3uzYbb7Y&t=429s&pp=ygUUYW5kcmV3IGhhbW1lbCByYXlsaWI%3D
Source 4	https://www.youtube.com/watch?v=xWWqhQ1JnvE&t=902s&pp=ygUUYW5kcmV3IGhhbW1lbCByYXlsaWI%3D
Source 5	https://www.youtube.com/watch?v=weYACOm6d8Y&t=244s&pp=ygUUYW5kcmV3IGhhbW1lbCByYXlsaWI%3D
Source 6	https://www.youtube.com/watch?v=MIpBqMBxJ8Y&pp=ygUUYW5kcmV3IGhhbW1lbCByYXlsaWI%3D
Source 7	https://stackoverflow.com/
Source 8	https://www.reddit.com/
Source 9	https://www.canva.com/
Source 10	@heykiyou (from Canva)

References for all Project

Source 11	@lokalldesign (from Canva)
Source 12	@gstudioimagen2 (from Canva)
Source 13	@graphicsrf (from Canva)
Source 14	@shenstock (from Canva)
Source 15	@rezual-karim-siams-images (from Canva)
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Source 29	https://www.raylib.com/
Source 30	https://github.com/raysan5/raylib